

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Master of Science in Information Technology (M.Sc.(IT))
Semester-I University Examination November 2019
MS 111 Object Oriented Programming in Java

Date: 14.11.19, Thursday

Time: 10.00 a.m. to 01.00 p.m.

Maximum Marks: 70

Instructions:

1. Attempt both sections in separate answer books.
2. Figure to the right indicates marks.
3. Make a suitable assumption when necessary.

SECTION– I

Q1 Do as directed.

[A] Select the output of the following java statements. **[03]**

1. `System.out.println('A'+ 'B');`
a. 131 b. 195
2. `int a[] = {1, 2, 3, 4, 5};`
`System.out.println(a.length);`
a. 6 b. 5
3. `System.out.println(Character.isDigit('a'));`
a. true b. false

[B] State True/False. **[05]**

1. Java is a weakly typed language.
2. The code generated by a java compiler is known as the byte-code.
3. Constructor is used to initialize the state of an object.
4. In java the word *true* is a Boolean literal.
5. An interface can extend from one or more interfaces.

[C] Fill in the blanks. **[03]**

1. _____ command is used to compile the java program.
2. Wrapping up (or binding) of data and functions together into a single unit (called class) is known as _____.
3. An _____ is instance of a class.

Q2 Answer the following questions.

[A] List the main concepts of OOP. Explain any five concepts briefly. **[06]**

[B] What is inheritance? List and explain the types of inheritance. **[06]**

OR

Q2 Answer the following questions.

[A] Explain the concept of method overloading and method overriding with example. **[06]**

[B] What is the use of package keyword? In how many ways one can use members of a class which is located in the other package? **[06]**

Q3 Answer the following questions.(Any Three)**[12]**

[A] Explain any four features of Java language.

[B] Explain the use of *this* keyword.

[C] Differentiate between an Abstract class and Interface.

[D] What is Boxing and Unboxing? Explain with suitable example.

[E] List the access specifiers used in java. Explain their use with suitable examples.

SECTION – II

Q4 Do as directed.

[A] Select the output of the following java statements. [03]

1. `int a[]={3,7,10,16}; System.out.println(a[10]);`
a. `ArithmeticException` b. `ArrayIndexOutOfBoundsException`
2. `System.out.println("JAVA".indexOf('J'));`
a. 0 b. 1
3. `System.out.println(10|5);`
a. 15 b. 14

[B] State True/False. [05]

1. Modulus operator (%) can be applied to Integer as well as Float values.
2. The object cloning is a way to create exact copy of an object.
3. The *static* keyword in java is used for memory management mainly.
4. The *join()* method of Thread is used to stop executing a thread for the specified amount of time.
5. *friendly* is one of the modifiers available in java.

[C] Fill in the blanks. [03]

1. JVM stands for _____.
2. ArrayList is available in java. _____ package.
3. _____ keyword is used to derive a child class from a parent class in inheritance.

Q5 Answer the following questions.

[A] What is Exception handling? Explain try, catch and finally block with example. [06]

[B] What are java threads? Explain the life cycle of thread. [06]

OR

Q5 Answer the following questions.

[A] How many ways thread can be created in java? Explain with example. [06]

[B] Write a code to read a binary file and display its content. [06]

Q6 Answer the following questions. (Any Three)

[12]

- [A] Explain Iterator interface with example.
- [B] Show the input of various types of data using Scanner class.
- [C] Explain the use of *throws* clause with example.
- [D] Write a code to display "Hello students" using an applet.
- [E] Explain the terms Component, Container and Frame.
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